



ICT Integration in the Learning Process



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Session Outcomes

By the end of the session, the participant should be able to:

- a) State the meaning of ICT integration in the learning process**
- b) Identify ICT resources that can be used during an ICT integrated lesson**
- c) Identify factors that influence integration of ICT in a lesson.**



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Activity 1: What is ICT integration in education?

Step 1: Organise the participants into manageable groups

Step 2: In their groups, allow them to brainstorm on the meaning of ICT integration in the learning process.

Step 3: Allow them to present your responses in plenary



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Definition of ICT integration in education

- ICT integration is the use of ICTs during the learning process to enhance the attainment of learning outcomes and appropriate competencies, namely, knowledge, skills, attitudes and values.



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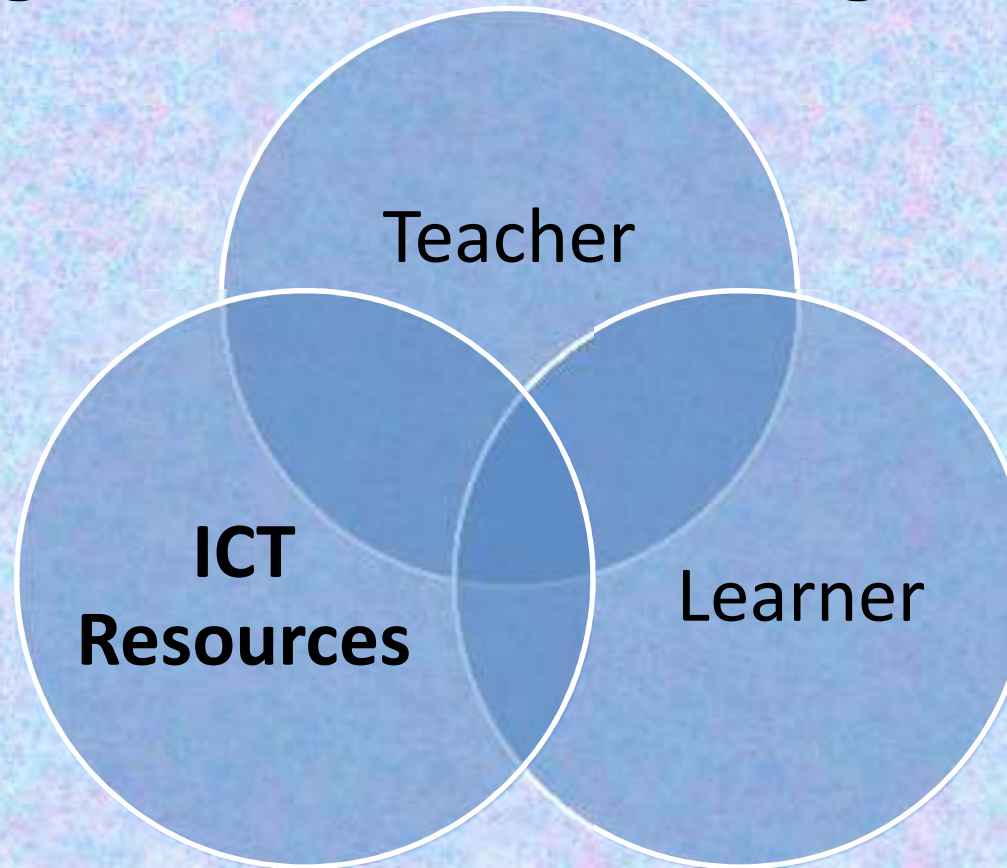
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What is ICT Integration?

ICT Integration in the learning





Activity 2: Why ICT integration?

Step 1: Organise participants into manageable groups.

Step 2: Allow them to discuss why we need to integrate ICT in the learning process.

Step 3: Ask the participants to present your responses during plenary.



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Possible answers

- We are already living in the 4th Industrial Revolution (4IR).
Mobile banking, self-driving cars, e-assessment among others are taking the centre stage.
- We are living in the era of Artificial Intelligence, Internet of Things and Big data (Global debates are revolving around here)
- Need to equip our learners with 21st Century Skills.



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Possible answers

- ICTs assist in simplifying abstract concepts
- ICTs arouse the curiosity of the learner
- ICTs increase the retention of the knowledge acquired
- ICTs enable learners acquire digital literacy skills
- ICTs make learning fun
- ICTs increase creativity of the learner.

(Any other relevant answer)



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Activity 3: ICT tools used during the learning process

Ask participants to discuss in groups some of the ICT tools a teacher can use during the learning process.

- Give an example of how each can be integrated in the learning process. Base your examples on a specific learning area from the Grade 4 designs.
- Ask them to present their findings during plenary.





Possible answers:

ICT tools that can be used during the learning process:

- Phone
- Camera
- Computer
- Laptop
- Projector
- Radio and TV - **KICD radio prog.s - KBC and on KEC**
www.kicd.ac.ke
- Tablets - **DLP devices** with content
- DVDs/CDs
- Flash discs
- Memory cards
- External hard discs
- Data cables (*Any other relevant answer*)



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NOTE:

While selecting ICT tools for use during teaching, it is advisable to find out if you have **learners with special needs** in your classroom.



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Activity 4: Adaptations done to take care of Learners With Disabilities

- Ask participants to identify the different categories of learners with special needs in their groups. How can ICT help make learning inclusive? Which adaptation strategies could help learners with special needs learn at the same pace with regular learners?





Possible answers:

Special need	Possible adaptations
Hearing	Headphones with amplified sound, signed videos
Visual	Narrations, braille adapted keyboard, Use of JAWS & NVDA
Physical	Voice input, adapted computer desk





Online Digital resources

Resource	Use
Digital Oxford University Dictionary	Checking out English vocaburary
Khan Academy	Maths and Science (Interactive videos)
PhET simulations	STEM based simulations
Kiwix	Offline version of Wikipedia



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Activity 4: In your groups, discuss the factors that need to be considered when selecting ICT resources to use in a lesson.

- Ask participants to share their points during plenary sessions



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Possible responses

- Relevance
- Ability to cater for different learning styles
- Level of language used
- Ability to be adapted for learners with special needs
- Promotion of 21st C skills
- Availability of the required ICT equipment





TPACK Model



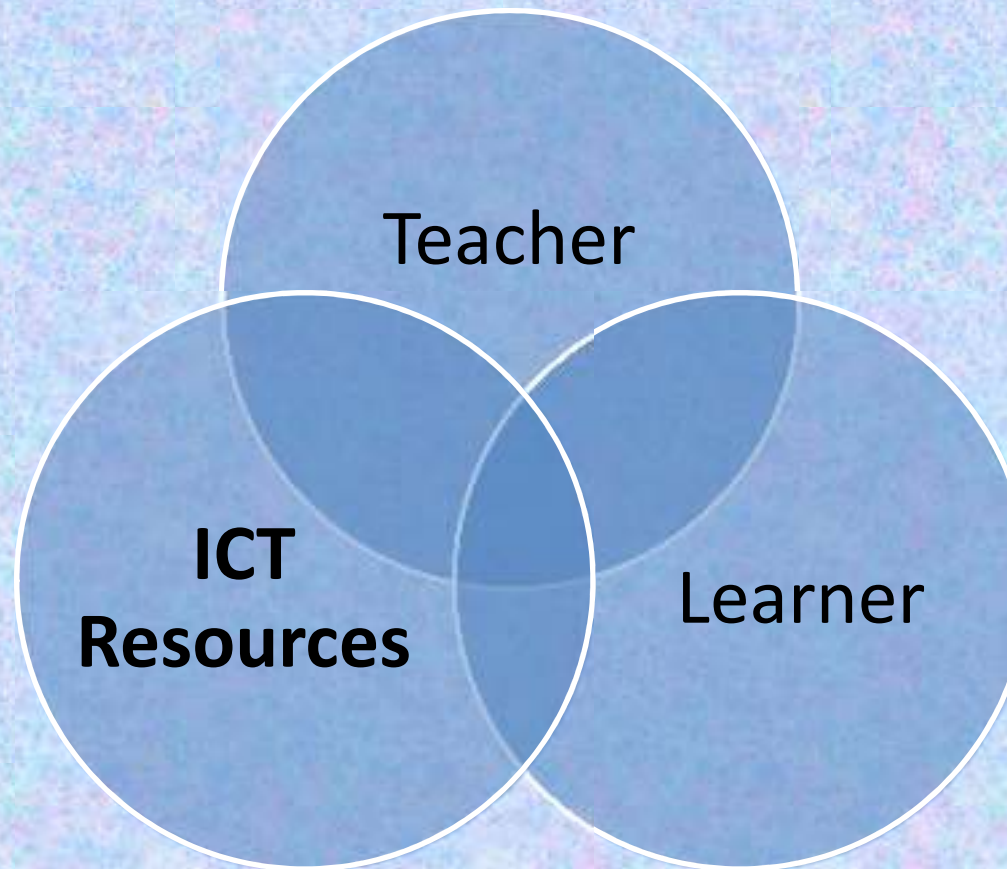
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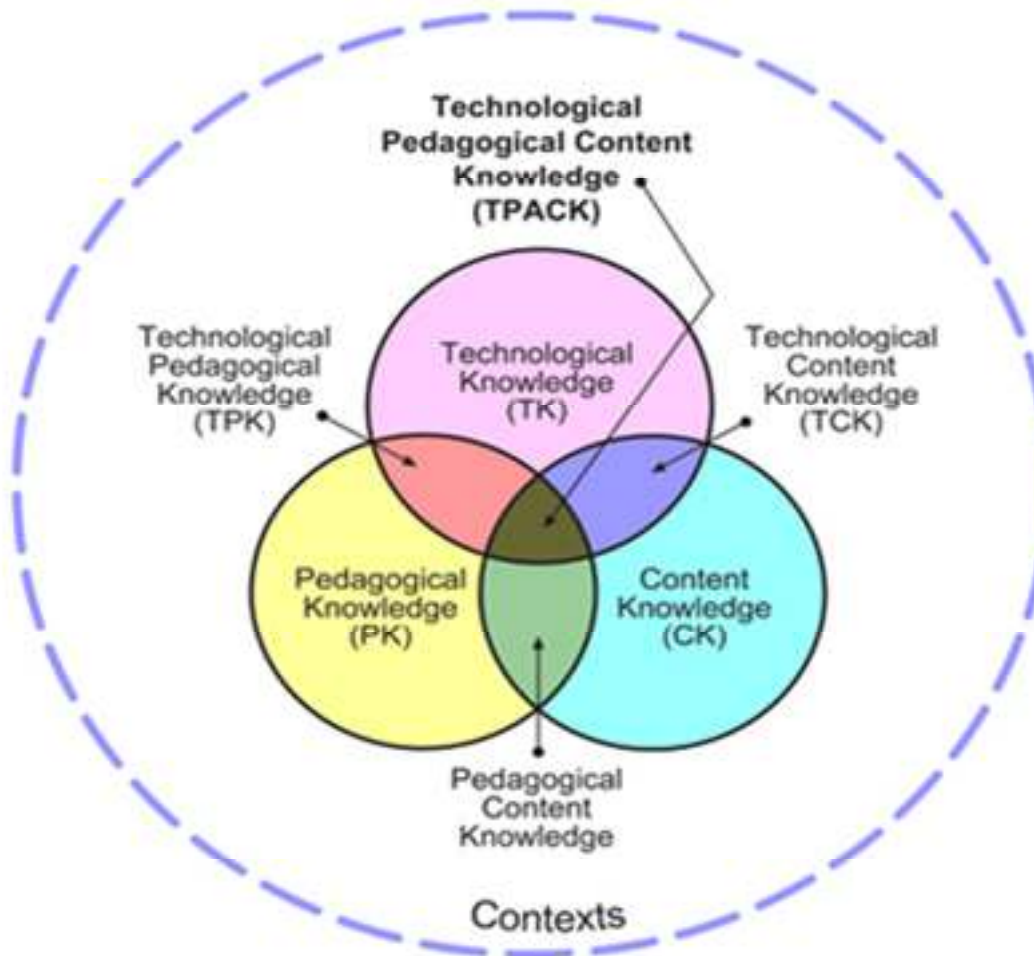


Key Players in the TPACK Model





Great TEACHING INVOLVES



Model 3: Technological Pedagogical Content Knowledge model (TPACK)





Factors influencing integration of ICT in teaching and learning

- **School leadership**

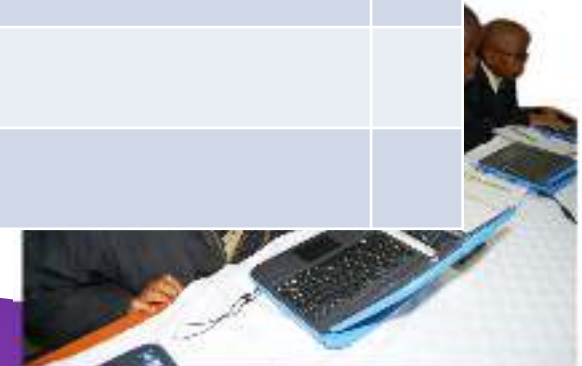
- Capacity building of teachers

- Time

- Availability of ICTs

- Access to ICTs

- Attitude





List of Abbreviations & Acronyms

DVD	Digital Versatile Disc
EAC	Ease of Access Centre
HDMI	High Definition Multimedia Interface
JAWS	Job Access With Speech
NVDA	Non Visual Desktop Access
VGA	Video Graphics Array





Thank you!

Q & A session



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